

14 A student investigated the hydrolysis of lipid in high-fat milk, using the enzyme lipase.

- 1 cm³ of enzyme solution was added to 10 cm³ of high-fat milk.
- The temperature was kept constant.
- The pH of the reaction mixture was recorded at time 0 minutes and every minute for 20 minutes.

Which statements correctly describe the expected results of this investigation?

- 1 The product forms more slowly as time proceeds because the concentration of the substrate is decreasing.
- 2 The pH of the reaction mixture increases rapidly in the first few minutes and then increases less rapidly.
- 3 The increase in the concentration of product eventually causes the lipase molecules to denature.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only